Hanji (Zhiheng) Xu
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Education

Rhode Island School of Design *Aug 2025 - May 2027*
MID Master of Industrial Design (Graduate Fellowship Recipient)

University of California, Berkeley *Aug 2021 - Aug 2025*
B.A. Cognitive Science (magna cum laude), Minors in Data Science and Landscape Architecture *GPA: 3.95*

Work in Design

Principal Material Researcher | **Softmode.ai, an LLM interface activated through interacting with clay** *Mar 2026 - Present*
- Led material research across a 5-person SWE/design/strategy team, exploring clay's physical properties as metaphor.
- Translated sensor data streams into text and image responses via the Claude API, computer vision, and Arduino.
- Won the Tenstorrent Sponsor Prize at MIT Media Lab Hard Mode Hackathon.
- Selected by Anthropic to exhibit at invitation-only Code with Claude conference in San Francisco, London, and Tokyo.

Inventor | **Universal Clock, for scheduling meeting time between time zones** *July 2024 - Present*
- Granted 2 Chinese patents (utility model CN 224889036 U; design CN 310257717 S); CNC-milled wood and 3D-printed plastic components through 40+ iterations.
- Led 7-person cross-functional team across engineering, business, and design.
- Delivered 2 production units with buyers praising quality and ease of international meeting coordination.
- Exhibited during Fidget Camp at Pebblebed VC, generating 32 purchase inquiries from 500+ visitors.

Principal Designer | **ida (hi-ida.com), a Compact Reusable Menstrual Cup Startup** *July 2023 - Present*
- Secured $23.5k grants, led a cross-functional team (US/China/Singapore) and medical advisors for FDA Class II device.
- Mapped out 5-year strategy and budget for patent filing, clinical trials, quality compliance, and supplier sourcing.
- Prototyped 4 iterations using processes including plaster casting, silicone and metal molding, and metal heat treatment.
- Built verification test fixtures, modeling the vaginal environment through CAD and in-house manufacturing.
- Validated market demand through 151 global user interviews; exhibited at Open Sauce Convention with 20k+ attendees.

Research Intern | **Morphing Matter Lab, UC Berkeley (NSF-funded E-Seed project)** *May 2025 - May 2026*
- Developed educational toolkits, exploring biomaterial properties to translate hygromorphic self-burial engineered seed research into K-12 outreach activities.
- Fabricated 4 educational modules using off-the-shelf parts with DFM-optimized, flat-packable design enabling replication across collaborating laboratories including CMU, UPenn, and Syracuse University, through documentation protocols.
- Assessed toolkit effectiveness through structured 60-minute user sessions and field research at educational exhibitions.

Landscape Architect | **2500 ft² Historic Berkeley Garden Redesign** *Jan 2024 - Apr 2024*
- Delivered 3 distinct design proposals (Japanese, modernist, native Californian) for a Julia Morgan property.
- Developed water-efficient planting around a heritage Deodar Cedar; produced site analysis, measurements, and plans.

Engineering Intern | **Shanghai Institute for Design & Research on Environmental Engineering** *June 2023 - July 2023*
- Developed architectural plans of a 150k ft^2 waste management facility for a Xianyou government contract.
- Drafted the 3D design with SketchUp and the 2D site plan with AutoCAD, Photoshop, and Illustrator.

Other Work

Operations Improvement Consultant | **Highlight Tech Corp. (TWSE: 6208)** *Aug 2026 - Sep 2026*
- Diagnosed process inefficiencies within 5 days at HTC, a vacuum system provider serving TSMC and Applied Materials.
- Deployed a zero-cost batch-generation fix that eliminated 80,000 manual edits/year (200 labor hours).
- Corrected valve leak-rate format in QC documentation to remove ambiguity in engineering requirements.

Summer Farm Laborer | **Brookview Orchard Limited** *Dec 2024 - Jan 2025*
- Removed suckers and weeds across 12,000 vines in 10-hectare first-season Zespri kiwifruit orchard in New Zealand.

Machine Learning Intern | **Energesque (Berkeley Skydeck Startup)** *May 2024 - Aug 2024*
- Trained KNN, Random Forest, XGBoost, and LSTM models for solar energy prediction; improved mean absolute error and computation time by 10x over the team's prior approach.
- Cleaned 500,000+ rows of meteorological data from NASA, Solcast, and OpenWeather and distilled into 2 suggestions.

Environmental Justice Researcher | **The Intersection of Homelessness and Flood Risk Management** *Jan 2024 - Aug 2024*
- Collected field data along 6 miles of Coyote Creek on infrastructure, encampments, and services; interviewed agencies.
- Won 1st place at ASFPM 2024 with an ArcGIS map documenting the overlap between floodplain and encampments.
- Presented at Valley Water's public hearing and solution summit.

Projects

Cyan Project | *A free summer camp for underprivileged students in Jiangxi, China* *July 2015 - July 2023*
- Co-founded and led a free summer sex-ed program in rural Jiangxi, China with 20 volunteers, reaching 300+ students.

Ergonomic Design Lead | *Drywall Transportation Device for Construction Workers* *Jan - May 2023*
- Designed shoulder-mount clip, reducing RULA score from 10 to 7 and improving NIOSH lifting index by 14.6%.
- Identified 42.7° neck strain and 140° elbow flexion during material handling tasks, and sketched 4 solution concepts.

- Prototyped ergonomic device based on the anthropometric analysis of 5th to 95th percentile male users.

Honors & Certifications

Accepted to demo at UIST 2026 Student Innovation Contest | *Built a telephone that senses bio signals.* *Nov 2026*
reharvest repair, Juried Group Exhibition | *Juci iPhone selected from 140+ entries; Providence Public Library.* *Oct - Dec 2026*
Utility Model Patent CN 224889036 U | *Multi-time-zone synchronized clock (ZL 2025 2 2720633.6), sole inventor.* *Oct 2026*
Design Patent CN 310257717 S | *Clock (ZL 2026 3 0301245.1), sole designer.* *Oct 2026*
Jingdezhen Taoxichuan Artist Residency | *2 Month ceramic phenomenological art.* *June 2026*
$880 Grant | *RISD ID Health Fund sponsored development of 4 hobbyist fume hoods for respiratory health.* *Mar 2026*
Top 30 Finalist | *Among 150 startups in the Berkeley Big Ideas Entrepreneurship Contest with ida.* *Jan 2024*
Millennium Fellowship Campus Director | *UN impact program with 0.3% acceptance rate for furthering 3 SDGs.* *Aug 2023*

Skills

Coding : Python, Java, SQL, NumPy, Scikit-learn, Pandas, CSS, HTML, JMP by SAS
Software : Rhino, Grasshopper, Keyshot, Blender, Adobe Suite, Figma, SketchUp, AutoCAD, Fusion 360, Solidworks, GIS
Prototyping : 3D Printing, CNC Mill and Lathe, Injection Molding, Power Carving, Chainsaw, Ceramics & Wood & Metal, Welding
Repairing : Juci (Staple Porcelain Restoration), Sewing
Communication : English, Mandarin, French (limited), Grant Writing, Public Speaking, Prior Art Research
Emergency : First Aid, Cardiopulmonary Resuscitation (CPR), Stop The Bleed (STB)

Interests

Trip Planning, Sleeping on public transit, Long-distance Biking and Running, Swimming, Snowboarding, Building Igloo, Jazz, Decluttering & Gifting, Conserving Utilities, Lock-picking, Pickling Kimchi